

CMSC 207: Chapter 3 Homework

March 9, 2026

1. Prove that the following two statements are not logically equivalent. In your proof, completely justify your answer.
 - a. A real number is less than 1 only if its reciprocal is greater than 1.
 - b. Having a reciprocal greater than 1 is a sufficient condition for a real number to be less than 1.
-

Let $P(x)$ be the statement $x < 1$, and let $Q(x)$ be the statement $\frac{1}{x} > 1$.

Statement (a) can be rewritten as “for any x , $P(x)$ only if $Q(x)$ ” which is equivalent to:

$$\forall x, \neg Q(x) \rightarrow \neg P(x).$$

Statement (b) can be rewritten as “for any x , $Q(x)$ is sufficient for $P(x)$ ” which is equivalent to:

$$\forall x, Q(x) \rightarrow P(x).$$

The conclusions of each of these conditional statements are inverses of each other, which are not logically equivalent. ■

Alternative Proof

Choose $x = -1 \in \mathbb{R}$. Then:

- $P(x)$ is the statement $-1 < 1$, which is true.
- $Q(x)$ is the statement $\frac{1}{-1} > 1$, which is false since $\frac{1}{-1} = -1 < 1$.

Since there exists an $x \in \mathbb{R}$ s.t. $P(x)$ is true but $Q(x)$ is false, the two statements are not logically equivalent. ■

2. Prove that the following is a valid argument:

All real numbers have nonnegative squares.

The number i has a negative square.

$\therefore$ The number i is not a real number.

Let $P(x)$ be the statement “ x is a real number”, and let $Q(x)$ be the statement “ x^2 is nonnegative”.

The argument can be rewritten as:

$\forall x, P(x) \rightarrow Q(x)$	given
$\neg Q(i)$	given
$\therefore \neg P(i)$	by universal modus tollens. ■